

Human Review: Linearizing Lipschitz Extension Operators

Original automatic proof: `solution_packet.pdf`

Checked date: 15 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears to correctly answer the question from Sofi's Remark 2.2(ii). The key point is that a Lipschitz, not necessarily linear, extension operator

$$F : \text{Lip}_0(Z) \longrightarrow \text{Lip}_0(X), \quad F(f)|_Z = f,$$

can be replaced by a bounded linear extension operator by averaging its finite differences over the additive abelian group $\text{Lip}_0(Z)$. Once this linearization is obtained, Sofi's Corollary 2.4 applies and gives that X is isomorphic to a Hilbert space.

The idea is sound. However, the notation in the automatic proof is slightly more abstract than necessary. Introducing temporary symbols such as Y and V makes the proof harder to read, especially at the averaging step. It is clearer to keep the spaces visible throughout: the domain group is $\text{Lip}_0(Z)$, and the values lie in $\text{Lip}_0(X) = \mathcal{F}(X)^*$.

Notation-Improved Verification of the Averaging Step

Assume, after replacing F by $F - F(0)$, that

$$F(0) = 0, \quad F(f)|_Z = f$$

for every $f \in \text{Lip}_0(Z)$. Let

$$L = \text{Lip}(F).$$

Since $\text{Lip}_0(Z)$ is an additive abelian group, it admits an invariant mean. Thus there is a positive norm-one functional

$$m : \ell_\infty(\text{Lip}_0(Z)) \longrightarrow \mathbb{R}$$

such that

$$m(\varphi(\cdot + a)) = m(\varphi) \quad (a \in \text{Lip}_0(Z), \varphi \in \ell_\infty(\text{Lip}_0(Z))).$$

For each $h \in \text{Lip}_0(Z)$, define the bounded $\text{Lip}_0(X)$ -valued function

$$\Delta_h : \text{Lip}_0(Z) \longrightarrow \text{Lip}_0(X)$$

by

$$\Delta_h(g) = F(g + h) - F(g), \quad g \in \text{Lip}_0(Z).$$

This is bounded because

$$\|\Delta_h(g)\|_{\text{Lip}} = \|F(g + h) - F(g)\|_{\text{Lip}} \leq L\|h\|_{\text{Lip}}$$

for every $g \in \text{Lip}_0(Z)$.

Now use the canonical duality

$$\text{Lip}_0(X) = \mathcal{F}(X)^*.$$

The invariant mean m only acts on scalar-valued bounded functions, so the average of Δ_h is defined weak-star coordinate by weak-star coordinate. Namely, define $G(h) \in \mathcal{F}(X)^* = \text{Lip}_0(X)$ by

$$\langle \mu, G(h) \rangle = m(g \mapsto \langle \mu, \Delta_h(g) \rangle), \quad \mu \in \mathcal{F}(X).$$

For fixed $\mu \in \mathcal{F}(X)$, the function

$$g \mapsto \langle \mu, \Delta_h(g) \rangle$$

is bounded, since

$$|\langle \mu, \Delta_h(g) \rangle| \leq \|\mu\| \|\Delta_h(g)\|_{\text{Lip}} \leq L \|h\|_{\text{Lip}} \|\mu\|.$$

Therefore the definition is legitimate. Moreover,

$$|\langle \mu, G(h) \rangle| \leq L \|h\|_{\text{Lip}} \|\mu\|,$$

and hence

$$\|G(h)\|_{\text{Lip}} \leq L \|h\|_{\text{Lip}}.$$

Linearity follows from translation invariance of the mean. Indeed, for $h, k \in \text{Lip}_0(Z)$,

$$\Delta_{h+k}(g) = F(g + h + k) - F(g) = \Delta_k(g + h) + \Delta_h(g).$$

Averaging over g , the first term satisfies

$$m(g \mapsto \langle \mu, \Delta_k(g + h) \rangle) = m(g \mapsto \langle \mu, \Delta_k(g) \rangle)$$

by invariance of m . Thus

$$G(h + k) = G(h) + G(k).$$

Since G is additive and satisfies

$$\|G(h)\|_{\text{Lip}} \leq L \|h\|_{\text{Lip}},$$

it is continuous at the origin. Additivity gives $G(qh) = qG(h)$ for rational q , and continuity then extends this identity to every real scalar. Hence G is real linear.

Finally, G is still an extension operator. Let

$$\rho : \text{Lip}_0(X) \longrightarrow \text{Lip}_0(Z)$$

be restriction. If

$$J : \mathcal{F}(Z) \longrightarrow \mathcal{F}(X)$$

is the canonical map induced by the inclusion $Z \subset X$, then

$$\rho = J^*.$$

For $\nu \in \mathcal{F}(Z)$,

$$\begin{aligned} \langle \nu, \rho G(h) \rangle &= \langle J\nu, G(h) \rangle \\ &= m(g \mapsto \langle J\nu, F(g + h) - F(g) \rangle) \\ &= m(g \mapsto \langle \nu, \rho F(g + h) - \rho F(g) \rangle) \\ &= m(g \mapsto \langle \nu, (g + h) - g \rangle) \\ &= \langle \nu, h \rangle. \end{aligned}$$

Therefore

$$\rho G(h) = h,$$

so $G(h)|_Z = h$. This proves that the original Lipschitz extension operator can be linearized into a bounded linear extension operator.

Review Outcome

Archive this packet as an apparently correct full solution, pending any later author or literature feedback. The proof's central averaging argument looks correct; in a polished version, I would recommend using the notation above rather than the more compressed Y, V notation, since it makes clear that Δ_h is a bounded $\text{Lip}_0(X)$ -valued function on $\text{Lip}_0(Z)$, and that the invariant mean is applied only after pairing with elements of $\mathcal{F}(X)$.